

Indic-Classify: Kannada News Headline Categorization with Supervised Deep Learning Models

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Abstract—This project focuses on developing a deep learning-based Kannada news headline categorization system to address the lack of NLP solutions for regional Indian languages, where challenges like script variations, compound words, and morphological complexity limit classifier performance. Using a dataset of ~120,000 labelled sentences collected from Kannada news portals across 10+ categories (e.g., Politics, Sports, Technology, Health), the work builds upon prior research that demonstrated strong results with pretrained Indic models (mBERT, XLM-RoBERTa) and deep learning architectures (Bi-LSTM, GRU, CNN), while noting issues like data imbalance and interpretability. The proposed approach combines traditional NLP feature engineering with modern transfer learning and ensemble methods, evaluating models through metrics such as accuracy, recall, precision, and F1-score. Implementation will leverage Python, PyTorch, HuggingFace Transformers, and IndicNLP, with deployment planned via a Gradio web application containerized in Docker. The solution aims to bridge the research gap in Indian language NLP, enhance inclusivity, and enable applications like news categorization, and recommendation systems.

Keywords— *Kannada News Classification, Deep Learning, XLM-R, mBERT, DistilMBERT, Text Categorization, Supervised Learning, Natural Language Processing*

I. INTRODUCTION

India's internet is buzzing with 400+ million users engaging in regional languages like Kannada, creating an urgent need for smart, automated content management systems [1]. Sorting news articles manually is a slow, costly, and error-prone task, which makes it tough to deliver real-time news updates or tailor content to individual readers' preferences [2].

The Indic-Classify project is tackling this challenge head-on by building a cutting-edge deep learning system to classify Kannada news headlines. The aim is to create a fast, reliable, and scalable tool that streamlines content workflows, making it easier for categorization of Kannada news headlines. Working with Kannada, a language known for its rich grammar and complex word structures, is no small feat, especially since it lacks large, well-organized digital datasets [3]. The project rises to this challenge by carefully collecting and curating high-quality data, resulting in a robust classifier and a valuable, standardized dataset that other researchers can

build on [4]. By adapting and fine-tuning advanced AI models like transformers, the system is designed to grasp the unique linguistic nuances of Kannada, ensuring accurate and context-aware headline classification [5]. This initiative not only enhances content accessibility but also paves the way for future advancements in regional language processing [6].

II. KANNADA NEWS HEADLINE CATEGORIZATION USING SUPERVISED DEEP LEARNING MODELS

The Indic-Classify project used three powerful, pre-trained AI models to classify Kannada news headlines. These models, known as transformer-based models, were chosen because they're great at understanding multiple languages, which is a big help for a language like Kannada that doesn't have a lot of digital resources [7]. Here's a breakdown of the models, how they work, and why they were picked:

A. The Models and How They Work

1) XLM-Roberta (XLM-R)

XLM-Roberta (XLM-R) is a multilingual transformer-based AI model trained on over 100 languages, including Kannada, making it highly effective for news categorization tasks [8]. It processes Kannada headlines by breaking them into tokens using an Autotokenizer, converting them into numerical representations, and analyzing word relationships across multiple layers [9]. The [CLS] token used to capture the overall meaning of a headline and predict its category, such as Politics or Sports [10]. XLM-R was chosen because it can transfer knowledge from high-resource languages to Kannada and handle mixed-language text, ensuring accurate and reliable classification [11]. The architecture and workflow of the XLM-R model used in this project are illustrated in Fig. 1.

2) Multilingual BERT (mBERT)

Multilingual BERT (mBERT) is a transformer-based model trained on Wikipedia text over 100 languages, including Kannada, with 12 layers and 12 attention heads to capture different aspects of sentence meaning [12]. It processes Kannada headlines by breaking them into smaller word pieces, converting them into numerical form, and

analyzing their context across multiple layers, using the [CLS] token to summarize the headline and predict its category [13]. Chosen for its ability to transfer knowledge from other languages, mBERT serves as a reliable baseline for classifying Kannada headlines, especially when data is limited [14]. The architecture and functioning of the mBERT model in this project are illustrated in Fig. 2.

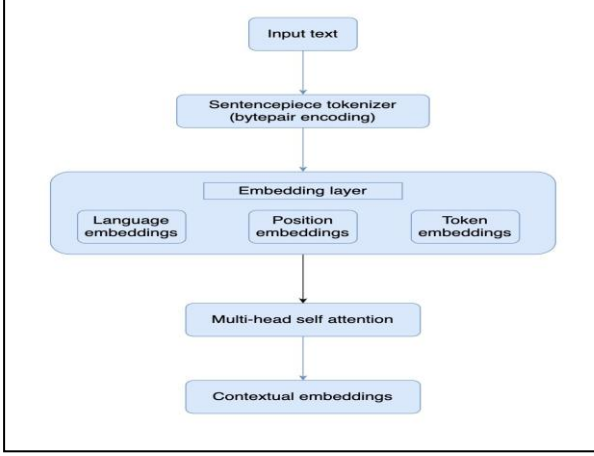


Fig. 1. XLM-R

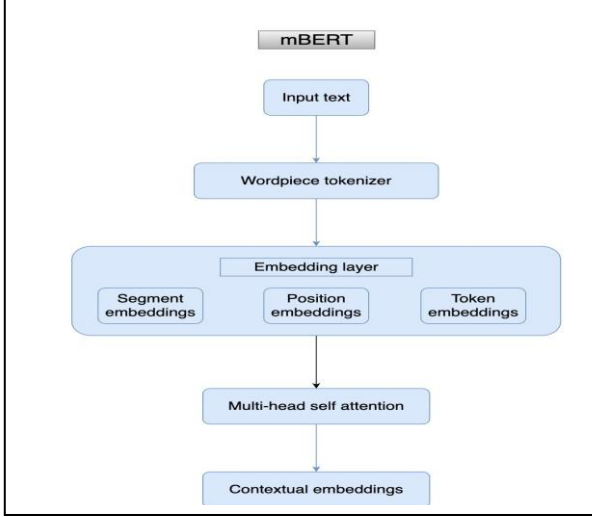


Fig. 2. mBERT

3) Distilled Multilingual BERT (DistilMBERT)

Distilled Multilingual BERT (DistilMBERT) is a compact and faster version of mBERT, designed with only 6 layers instead of 12, making it more memory-efficient while maintaining performance close to the original [15]. It processes Kannada headlines in the same way as mBERT—tokenizing text, converting it into numerical embeddings, and passing it through its layers before using the output to classify the headline's category [16]. Chosen for its speed and efficiency, DistilMBERT is useful in real-world applications with limited computing resources, such as deployment on websites, without sacrificing much accuracy [17]. The architecture and workflow of the DistilMBERT model used in this project are illustrated in Fig. 3.

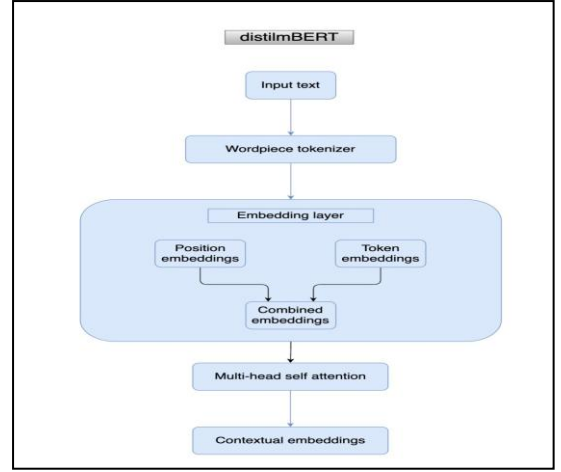


Fig. 3. DistilMBERT

III. METHODOLOGY

The Indic-Classify project followed a structured multi-step process to train and refine its AI models [18]. In Step 1 (Data Collection), we gathered a large dataset of Kannada news headlines across multiple categories [19]. In Step 2 (Preprocessing and Exploratory Data Analysis), the data was cleaned, tokenized, and normalized, while also being analyzed to understand patterns, category distribution, and potential challenges like imbalance [20]. In Step 3 (Initial Training), baseline deep learning models were trained to classify the headlines [21]. Finally, in Step 4 (Advanced Fine-Tuning), transformer-based models were fine-tuned and optimized for higher accuracy, ensuring robust performance on Kannada headline classification [22].

A. Data Collection

A corpus of 1,20,000 Kannada news headlines was collected from various sources like Prajavani and Vijaya Kannada [23], with 63 initial categories consolidated into 10 major classes such as Politics, Sports, and Entertainment to avoid data sparsity, as illustrated in Fig. 4 [24]. A preprocessing pipeline cleaned the text by removing special characters, HTML tags, and non-Kannada scripts as summarized in Table I, while Exploratory Data Analysis (EDA) examined distribution, class imbalance, and structural features like headline length [25]. Baseline feature extraction using TF-IDF, along with dimensionality reduction via PCA and t-SNE, confirmed the chances for clear category separation [26].

Smallest headline from each category	
Category	Headline
ರಾಜಕೀಯ	ಅಂತರಿಕ ಕಚ್ಚಾಟಿ ಮರಮಾಚಲು ಕಾಂಗ್ರೆಸ್ ಸುಳ್ಳು ಭರವಸೆ ನೀಡುತ್ತಿದೆ: ಅರುಣ್ ಸಿಂಗ್
ವಿದೇಶ	ಭಾರತ - ಪಾಕಿಸ್ತಾನ ನಡುವೆ ಪರಮಾಣು ಯುದ್ಧ ನಡೆದರೆ 125 ಮಿಲಿಯನ್ ಮಂದಿ ಸಾವು: ಅಧ್ಯಯನ
ಕ್ರೀಡೆ	MS Dhoni's ₹100-crore defamation case: 10 ವರ್ಷಗಳ ನಂತರ ಕೊನೆಗೊ ವಿಚಾರಣೆ ಕೈಗೊಂಡ ಮದ್ರಾಸ್ ಹೈಕೋರ್ಟ್
ವಾಣಿಜ್ಯ	ಕಳೆದ ನಗರೀಕರಣದಿಂದ 2050ಕ್ಕೆ 120 ಲಕ್ಷ ಕೋಟಿ ರೂ. ಹೊರ ಮನರಂಜನೆ
ಮನರಂಜನೆ	'ಸೈ ರಾ' ಕನ್ನಡ ಟೀಲರ್ ರಿಲೀಸ್: ಸುದೀಪ್ ಧ್ವನಿ ಕೇಳಿ ಫಿಲ್ಮ್ ಆದ ಅಭಿಮಾನಿಗಳು
ಅಪರಾಧ ಸುದ್ದಿ	ಎಸ್‌ಎಸ್‌ಎಲ್‌ಸಿಯಲ್ಲಿ ಮೂರು ಬಾರಿ ಫೇಲ್; ಮನನೋಂದ ಬಾಲಕನಿಂದ ದೇವರ ವಿಗ್ರಹ ವಿರೂಪ!
ಆರೋಗ್ಯ	ಈ 7 ಬಗೆಯ ಸ್ಪೃಲಿಂಟ್ ಕಿಲ್ಲರ್ ಕಾಯಿಲೆಗಳ ಲಕ್ಷಣಗಳನ್ನು ಕಣ್ಣುಗಳಲ್ಲಿ ನೋಡಿಯೇ ಪತ್ತೆ ಹಚ್ಚಬಹುದಂತೆ!
ಕೃಷಿ	ಭತ್ತಕ್ಕಿಂತ ಪರಿಸರ ಸ್ನೇಹಿ ವಿಧಾನಕ್ಕೆ ಮೂರ
ತಂತ್ರಜ್ಞಾನ	TikTok ಟಿಕಟಾಕ್ ಮೇಲಿನ ನಿಷೇಧ ತೆರವುಗೊಳಿಸಿಲ್ಲ: ಕೇಂದ್ರ ಸರ್ಕಾರ
ಸಾಹಿತ್ಯ	'ಮೂಡಲಪಾಯ ಯಕ್ಷಗಾನ' ಕಲೆಗೆ ಜೀವ ತುಂಬುವ ಚಿಣ್ಣು!

Fig. 4. Categories

TABLE I. PREPROCESSING STEPS

Preprocessing Steps	Description
HTML tag Removal	All HTML tags, scripts, and CSS content were programmatically stripped from the text to isolate the news headlines.
Special Character & Punctuation Filtering	Non-alphanumeric characters, symbols, and excessive punctuation were removed or replaced to reduce noise in the data.
Non-Kannada Character Removal	The data was filtered to remove any non-Kannada scripts or characters, ensuring the corpus is solely in the target language.
Whitespace Normalization	Multiple spaces and other forms of extraneous whitespace were consolidated into a single space to standardize the text format.
Consistent Formatting	The text was converted to a consistent format, such as lower-case, to minimize variations in the same word.

B. Preparing the Data and Training Process

- Category filtering: Initially, the data contained ~1,20,000 samples 63 categories, out of which 10 were chosen namely, Literature, Technology, Politics, International, Sports, Crime, Economics, Entertainment, Health and Agriculture [27]. The categories similar to the chosen ones were merged [28], like cinema was merged to Entertainment, cricket was to Sports, etc. All the other samples categories with no similarity were removed. After this step, we ended with a clean dataset with ~90,000 samples, each headline categorized into one of the 10 categories [29].
- Balancing the Data: Some categories, like Literature and Entertainment, had way more headlines than others as seen in Fig. 5, which could skew the models' performance [30]. To fix this, we removed some more samples from the categories having a large number of samples to minimise the skewness of the data [31]. The final dataset had ~60,000 samples.

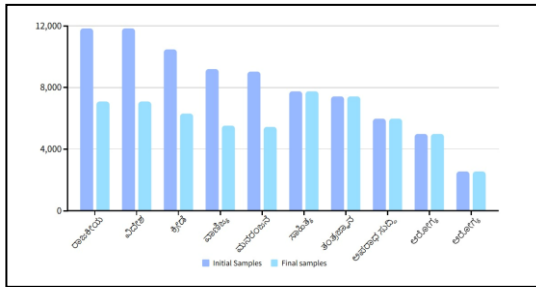


Fig. 5. Before and After Preprocessing

- Splitting the Data: The data was divided them into 80% for training (48,000 headlines) and 20% for testing (12,000 headlines) [32].
- Text normalization: We tidied up the headlines by removing punctuation, special symbols, and extra spaces to make the text clean and consistent for the models to process [33].

- Tokenization: The headlines were broken into smaller tokens using a tool called Autotokenizer, which works for all three models: XLM-R, mBERT, and DistilMBERT. This step helps the models understand the text better as shown in Fig. 6 [34]

	headline	category	headline_length	preprocessed_headline
0	ಕರ್ನಾಟಕ ಸರ್ಕಾರದ ಮುಖ್ಯಮಂತ್ರಿ ಸಿದ್ದರಾಮಯ್ಯ...	ರಾಜಕೀಯ	69	ಕರ್ನಾಟಕ ಸರ್ಕಾರದ ಮುಖ್ಯಮಂತ್ರಿ ಸಿದ್ದರಾಮಯ್ಯ...
1	ಬಳ್ಳಾರಿ ಜಿಲ್ಲೆಯಲ್ಲಿ ಉದ್ದೇಶಪೂರ್ವಕವಾಗಿ...	ರಾಜಕೀಯ	89	ಬಳ್ಳಾರಿ ಜಿಲ್ಲೆಯಲ್ಲಿ ಉದ್ದೇಶಪೂರ್ವಕವಾಗಿ...
2	ಕೊಪ್ಪಳ ಜಿಲ್ಲೆಯಲ್ಲಿ ಪಾಲಿಕಾರ್ಡ್ ಹಂಚಿಕೆ...	ರಾಜಕೀಯ	67	ಕೊಪ್ಪಳ ಜಿಲ್ಲೆಯಲ್ಲಿ ಪಾಲಿಕಾರ್ಡ್ ಹಂಚಿಕೆ...
3	ಕರ್ನಾಟಕ ಸರ್ಕಾರದ 2023-24ರ ಬಜೆಟ್...	ರಾಜಕೀಯ	80	ಕರ್ನಾಟಕ ಸರ್ಕಾರದ 2023-24ರ ಬಜೆಟ್...
4	ಗ್ರಾಮ ಪಂಚಾಯತ್‌ನಲ್ಲಿ ಕುಟುಂಬದ...	ರಾಜಕೀಯ	74	ಗ್ರಾಮ ಪಂಚಾಯತ್‌ನಲ್ಲಿ ಕುಟುಂಬದ...

Fig. 6. Tokenization

C. Model Training and Optimization

The fine-tuning process adapted pre-trained transformer models to Kannada news classification through carefully optimized hyperparameters [35]. Key configuration parameters included:

- Learning Rate: A conservative rate of 2×10^{-5} with linear scheduling and warmup period prevented catastrophic forgetting while enabling effective adaptation to the target task [36].
- Training Setup: Models were trained for 5 epochs with batch size 32, using AdamW optimizer for improved generalization. Early stopping based on validation performance prevented overfitting [37].
- Regularization: Dropout and L2 weight decay techniques maintained model generalization capability during the fine-tuning process [38].

IV. RESULTS AND DISCUSSION

The Indic-Classify project evaluated its models using metrics such as Accuracy, Precision, Recall, and F1-Score to gain a clear understanding of their strengths and weaknesses in categorization of Kannada news headlines [39]. Accuracy provided an overall measure of correct predictions, while Precision and Recall highlighted how good the models handled specific categories. The F1-Score offered a balanced view by combining both Precision and Recall, making it especially useful for handling imbalanced categories [40].

As shown in Table II, fine-tuning consistently improved model performance, with XLM-Roberta achieving the highest scores across all metrics.

TABLE II. RESULTS

Model	Phase	Accuracy	Macro F1-Score	Precision	Recall
XLM-Roberta	Baseline	83.18%	0.8253	0.8282	0.8256
	Fine-Tuned	85.89%	0.8549	0.8500	0.8533
mBERT	Baseline	80.37%	0.7951	0.7989	0.7927
	Fine-Tuned	83.32%	0.8265	0.8288	0.8255
DistilMBERT	Baseline	75.12%	0.7390	0.7399	0.7397
	Fine-Tuned	82.19%	0.8154	0.8186	0.8130

The training loss curves presented in Figs. 7–9 demonstrate effective and stable model convergence. The corresponding ROC-AUC curves shown in Figs. 10–12 indicate strong discriminatory capability across all classes. Furthermore, the confusion matrices provided in Figs. 13–15 confirm robust classification performance on the test set, with the majority of predictions concentrated along the main diagonal.

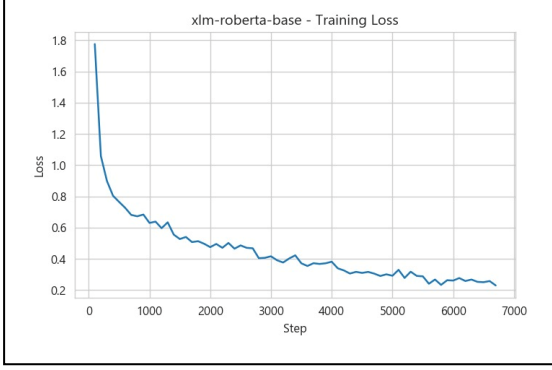


Fig. 7. Training loss curve for XLM-R

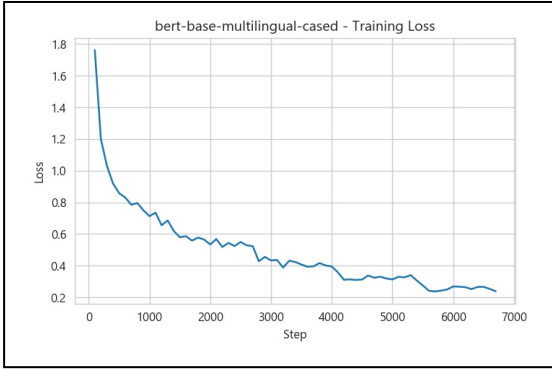


Fig. 8. Train loss curve for mBERT

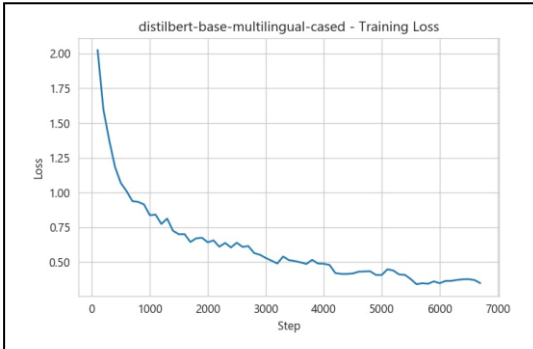


Fig. 9. Train loss curve for DistilMBERT

Our findings validate that fine-tuning transformer-based multilingual models, particularly XLM-R, is a highly effective approach for Kannada news headline classification [41]. With an accuracy of 85.89% and a macro-F1 score of 0.8549, our XLM-R model significantly outperformed mBERT (83.32%) and DistilMBERT (82.19%). These results surpass those of a comparable Kannada study that achieved only 65.71% accuracy using the same model as in [42], a success we attribute to our larger, more carefully curated dataset.

Our work also compares to prior efforts using traditional deep learning models. For instance, studies on Telugu news classification using architectures like Bi-LSTM reported maximum accuracies ranging from 63.67% to 70.92% as in [43]. The superior performance of our transformer models highlights their effectiveness in handling the complexities of regional Indian languages [44].

While some studies [45] achieved higher accuracies (up to 97%) using simpler, language-specific methods like N-gram models or specialized embeddings, our approach offers a critical advantage: generalizability. Multilingual transformers are designed for cross-lingual knowledge transfer, providing a better solution that can be applied to other low-resource languages with minimal retraining [46]. This positions our work as a scalable and practical foundation for developing future multilingual NLP systems [47].

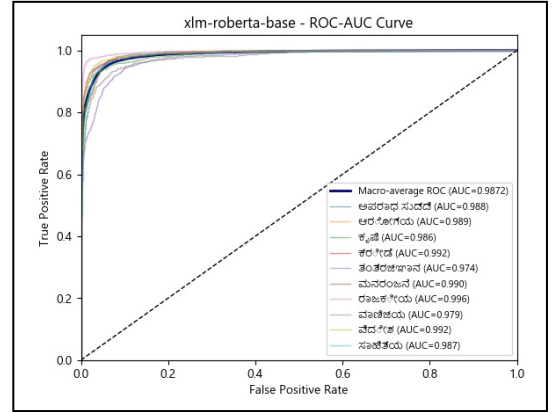


Fig. 10. ROC-AUC curve for XLM-R

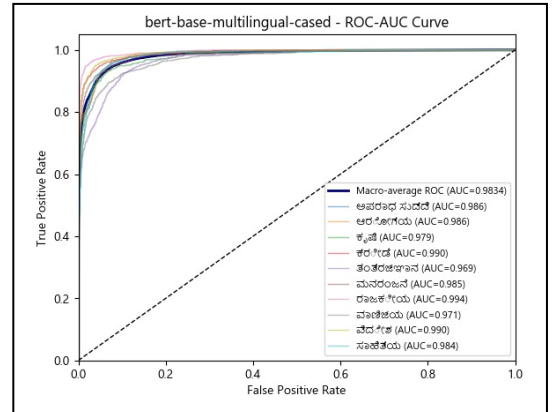


Fig. 11. ROC-AUC curve for mBERT

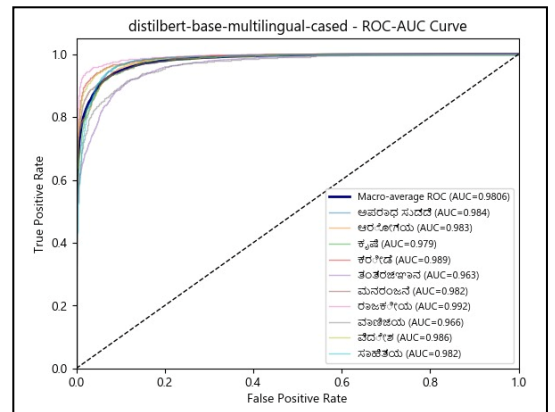


Fig. 12. ROC-AUC curve for DistilMBERT

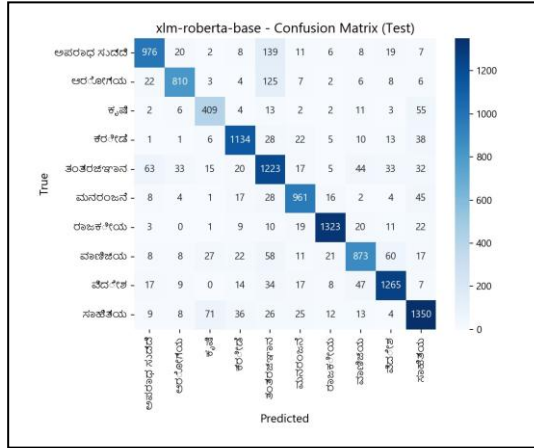


Fig. 13. Confusion matrix for XLM-R

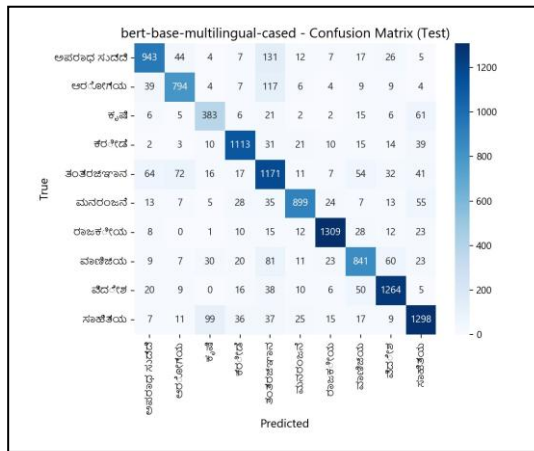


Fig. 14. Confusion matrix for mBERT

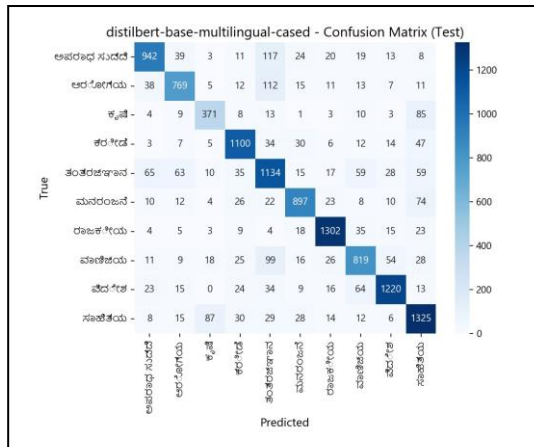


Fig. 15. Confusion matrix for DistilMBERT

V. CONCLUSION

The Indic-Classify project successfully addressed the challenge of Kannada news headline categorization by systematically progressing from data collection to model fine-tuning and evaluation. Using a dataset of 60,000 headlines, the study benchmarked multiple transformer-based models and achieved significant improvements, with XLM-R delivering the best performance (accuracy 85.89%, macro F1-score 0.8549), followed by mBERT and DistilMBERT. The results were validated through

comprehensive metrics, including accuracy, recall, precision, and F1-score, ensuring robustness and transparency. Overall, the project demonstrated the effectiveness of transformer architectures for low-resource languages like Kannada while providing reliable benchmarks for future research.

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